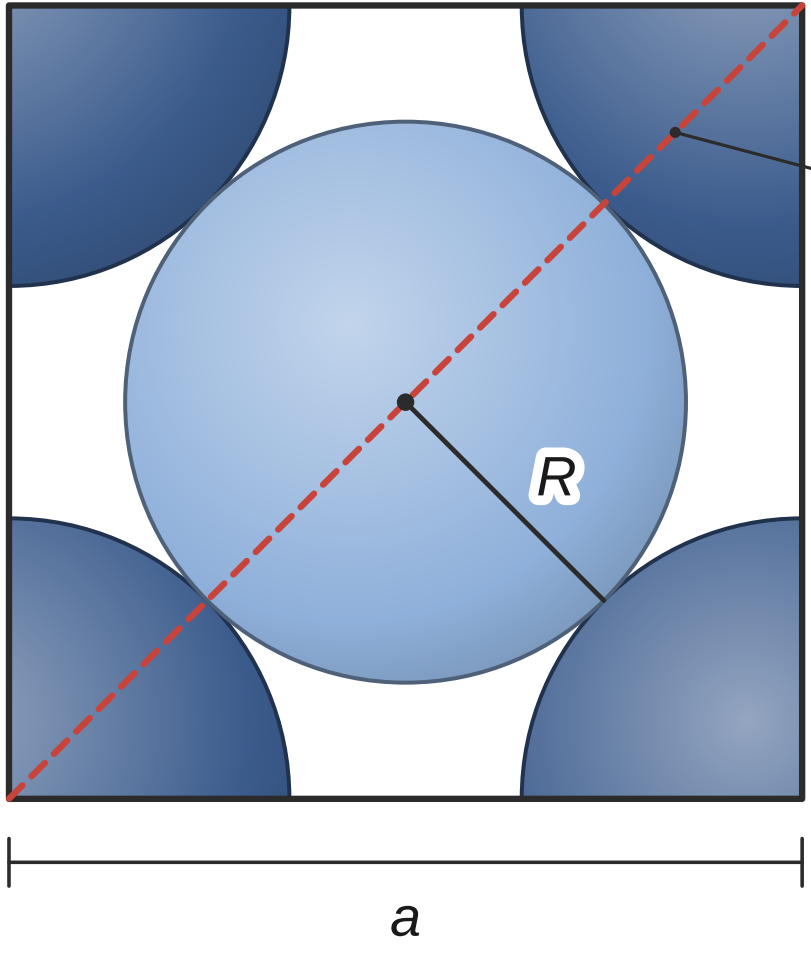

$a\sqrt{2} = 4R$

contact selon la diagonale de face